Stellenbosch University Faculty of Engineering

Module Framework

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This document should be read with the following documents:

- ***Stellenbosch University Calendar Parts 1 and 11.***
- ***Faculty of Engineering Assessment Rules***
- ***Faculty of Engineering General Stipulations for Undergraduate Modules[1]***

MECHANICAL PROJECT 478
MECHATRONIC PROJECT 478
MECHATRONIC PROJECT 488
STUDY GUIDE

2026

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[1] Available on SUNLearn for modules offered by Faculty of Engineering, in the block titled "General Programme Information" on the side of the screen

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DUE DATES FOR 2026 (12 MONTH SKRIPSIE PROJECTS)

Note: Attendance at all lectures is compulsory.

Fri 13 Feb	All students should have a topic.
Fri 13 Feb	12:00 (Room K302) Introductory lecture (F2F only), information regarding module, compulsory safety lecture.
Fri 20 Feb	Must complete the required online test for the safety lecture on SUNLearn. The test must be repeated until satisfactory marks are obtained.
16 - 20 Feb	Students should contact their supervisors to arrange their regular appointments (at least every 2nd week in the first semester and every week in the 2nd semester).
Fri 27 Feb	12:00 (Room K302) Lecture (F2F only) regarding project proposals. Slides will be made available on SUNLearn.
Fri 13 Mar	Hand-in written project proposal to supervisors. These must include the ECSA outcome self-assessment and a section on the responsible use of resources and end-of-life strategy.
Fri 8 May	12:00 (Room K302) Lecture (F2F only) regarding writing guide. Slides will be made available on SUNLearn.
Tue 17 Jun	15:00 Optional early hand-in date for final report (only for students wishing to complete the final year project in the first semester). No late hand-ins. All late submissions automatically roll over to a full year skripsie.
Wed 24 Jun	Oral examination only for first semester projects.
Winter recess	Work at least 100 hours on the Mechanical/Mechatronic Project.
Fri 25 Jul	Hand in progress report at your supervisor before 16:00.
Fri 31 Jul	11:00 (Room E3066) Lecture (F2F only) regarding Skripsie presentations. Slides will be made available on SUNLearn.
3 - 14 Aug	Project progress presentations (10-minute presentation per student – presentations via MSTeams).
Fri 21 Aug	Supervisors report on their students' progress to the course coordinator. If the progress is unsatisfactory, the project can be stopped.
Fri 21 Aug	(Room E3066) Lecture regarding project write-up (presentation by the Language Centre). Start time and format to be finalised.
Mon 5 Oct	Hand in a draft of the final report to the supervisor by 15:00.
Mon 5 Oct	After this date, no new submissions for manufacturing will be accepted by the M&M workshop without a written motivation by the supervisor.
Fri 16 Oct	Draft final report returned to students.
Fri 16 Oct	11:00 (Room E3066) Lecture (F2F only) regarding examination process, techno-economic analysis, report and posters. Slides will be made available on SUNLearn.
Fri 16 Oct	**16:00 All equipment must be returned to the store, and laboratories must be clean and neat. No further work on the project is allowed.**
Fri 23 Oct	**15:00 Electronic hand in via SUNLearn:**

Project file to supervisor (see 9.3 for content)
Single PDF document handed in via SUNLearn (see 9.3 for content). The following appendices must be included in the report:

- Outcome self-assessment
- Techno-economic analysis
- Risk analysis and safety procedures
- Responsible use of resources and end-of-life strategy

NB: Note that 5% per ¼ hour (or part thereof) will be subtracted for late submission, and the final cut-off time is at 17:00, after which no late submissions will be accepted. For further information regarding late submission or extension of the submission, please see 9.5.

9-20 Nov	Each student delivers a 10-minute presentation (see 9.4), followed by an oral examination by the supervisor and internal moderator. The external examiner is also present to question the student and to interact with the supervisor and internal moderator in order to finalise the student's mark. During this session, the student must also exhibit their poster (see 9.4), and must hand in all data (including the final report, project proposal, both oral presentations, poster, CAD models, data sheets, calculations, computer programmes, etc.) to their supervisor.
Mon 23 Nov	Students nominated by supervisors and internal moderators take part in an oral presentation competition for the SAIMechE award for the best and runner-up presentations.
Mon 23 Nov	Supervisor's nominations for prizes (including Jac van der Merwe competition).

DUE DATES FOR 2026 (6 MONTH SKRIPSIE PROJECTS)

Note: This schedule is a suggestion that can be altered in agreement with the supervisor to meet the needs of the project and the student. Also, students doing six-month projects are not required to attend the lectures. Attendance, where possible, is, however, encouraged.

Fri 13 Feb	All students should have a topic.
Fri 13 Feb	12:00 (Room K302) Introductory lecture (F2F only), information regarding module, compulsory safety lecture.
Fri 20 Feb	Must complete the required online test for the safety lecture on SUNLearn. The test must be repeated until satisfactory marks are obtained.
16 - 20 Feb	Students should contact their supervisors to arrange their regular appointments (suggested to be at least once a week).
Mon 2 Mar	Hand-in written project proposal to supervisors. These must include the ECSA outcome self-assessment and a section on the responsible use of resources and end-of-life strategy.
Mon 13 Apr	Hand in progress report to your supervisor before 16:00.
21 - 24 Apr	Project progress presentations (10-minute presentation per student – presentations via MSTeams). Six months students make individual arrangements with their supervisors for the progress presentations.
Tue 28 Apr	Supervisors report on their students' progress to the course coordinator. If the progress is unsatisfactory, the project can be stopped.
Mon 25 May	Hand in a draft of the final report to the supervisor by 15:00.
Fri 5 Jun	Draft final report returned to students.
Fri 12 Jun	**16:00 All equipment must be returned to the store, and laboratories must be clean and neat. No further work on the project is allowed.**
Wed 17 Jun	**15:00 Electronic hand-in via SUNLearn:** Project file to supervisor (see 9.3 for content) Single PDF document handed in via SUNLearn (see 9.3 for content). The following appendices must be included in the report: • Outcome self-assessment • Techno-economic analysis • Risk analysis and safety procedures • Responsible use of resources and end-of-life strategy

NB: Note that no late hand-ins will be accepted. If a student misses the hand-in deadline, the project automatically rolls over to a normal 12-month project.

Wed 24 Jun	Each student delivers a 10-minute presentation (see 9.4), followed by an oral examination by the supervisor and internal moderator. The external examiner is also present to question the student and to interact with the supervisor and internal moderator in order to finalise the student's mark. During this session, the student must also exhibit their poster (see 9.4), and must hand in all data (including the final report, project proposal, both oral presentations, poster, CAD models, data sheets, calculations, computer programmes, etc.) to their supervisor.

Stellenbosch University Faculty of Engineering

Module Framework

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MECHANICAL PROJECT 478 **MECHATRONIC PROJECT 478** **MECHATRONIC PROJECT 488** **2026**	**Lecturer(s):** Prof Martin Venter Tel: 808 4477, e-mail: mpventer@sun.ac.za **Admin:** Ms Chantel Maclons Tel: 808 4958, e-mail: chantelp@sun.ac.za	Approved by Programme Coordinator: Date: 02/2026

1 Assessment Details

- Main assessment dates and venues are provided at firga.sun.ac.za and my.sun.ac.za
- Method of assessment as indicated in the Calendar Part 11
- Note that awarding a pass mark is subject to meeting each of the ECSA Graduate Attributes assessed in this module, as stated in the Faculty of Engineering's Assessment Rules

Calculation of final marks:

Refer to section 12 for detail.

Continuation can at any stage be refused on the grounds of insufficient progress (see 9.2).

2 Notional Hours

- You should spend 10 notional hours per credit on this module over the course of the semester.
- It is envisaged that these hours will be allocated as follows:

Activity	Contact Hours	Self-study Hours
Lectures	6 x 0.83h (50 minutes each) = 5	0
Tutorials	0	0
Practicals	0	0
Assignments	0	435
Main assessments	0	10
Total (for this XX-credit module)	**450**	

3 Language of Tuition

- The language of tuition in this module is according to the Faculty's approved Language Implementation Plan. Please refer to the website of the Engineering Faculty or the "General Information" block on SUNLearn for the particulars.

4 Module Objectives

Aim: Independent execution of theoretical and/or practical design and/or investigation in the field of mechanical/mechatronic engineering with formal oral presentations and the submission of a final comprehensive report.

A student who has successfully completed this module can:

1) Solve engineering problems by:
   a) Applying a synthesis of knowledge and technologies from different basic and engineering sciences.

[1] Available on SUNLearn for modules offered by Faculty of Engineering, in the block titled "General Programme Information" on the side of the screen

b) Application of engineering methods, skills and techniques to arrive at practical results and sound assessment of results.
c) Work creatively by, amongst others, independently assembling and utilising new information.
d) Apply sound engineering judgement.

2) Independently and successfully complete a mechanical/mechatronic project within the stipulated due dates.
3) Plan a project, taking objective formulation, steps required and time schedules into account.
4) Compile technical reports on the planning, progress and the whole of a project.
5) Give professional oral presentations on the progress and the results of a project.
6) Demonstrate a grasp and appreciation of professional ethics and practice in the execution of an engineering project.

Prior knowledge required:
The module requires final-year enrolment.

Proceeding application:
This module is the end of the module chain. It is used in engineering practice in areas such as design, thermofluid dynamics, mechanics, electrical systems, and computer systems.

5 Module Content and Schedule

Prescribed textbook(s): No prescribed books

Notes: Guidelines for Project Proposals
Guidelines for Technical Presentations
Guide for Writing Technical Reports
Guidelines for Poster Presentations
Notes on techno-economic analysis
Notes on responsible use of resources and end-of-life strategy
Notes on safety with laboratory setups
Any other sources indicated by the module coordinator
*** All notes are made available on the SUNLearn page for this module*

Other sources: The supervisor may indicate additional sources, as dictated by the project.

Period and Duration	Subject
Fourth term of the preceding year	Students can propose projects to their supervisor. Lecturers compile a list of potential projects.
First semester of fourth year 2.5 hr per week = 30 hr	Each student chooses projects and indicates their order of preference. Projects are assigned to students. Lectures (published online on SUNLearn) are presented on professional oral presentations and written communication. Students must compile a written project proposal. Students must start working on their projects.
Winter recess	Students work on their projects. The winter recess should not be used to complete Vacation Work.
Second semester of fourth year 35 hr per week = 420 hr	Lectures (published online on SUNLearn) are presented on professional oral presentations and written communication. Students must do an oral presentation on the progress of the project. Complete the work on the project. Students complete a final report.
November examination period	Students must each deliver an oral and poster presentation. All final reports are assessed by the supervisors, internal moderators and external examiners.

6 ECSA Knowledge Area Credits

Mathematical Sciences	Natural Sciences	Engineering Sciences	Design and Synthesis	Complementary Studies
0	0	0	39	6

Design and Synthesis: Solving a complex engineering problem. This involves, amongst other things, the design of components, systems, engineering works, products and processes. The solution involves a synthesis of knowledge from senior-level modules as well as new knowledge.

Complementary Studies**:** By means of formal lectures, students learn how to write a comprehensive project report, how to report on the project using an oral presentation and also how to convey the essence of the project to the general public by means of a poster.

7 ECSA Graduate Attributes

GA 1. Problem solving: Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.	
How is the Outcome Assessed?	The candidate must successfully complete an individual project. The project is assessed using a number of reports, oral presentations and poster as stipulated in the due dates section at the start of this document. The candidate applies in a number of varied instances, a systematic problem solving method including: 1. Analyses and defines the problem, identifies the criteria for an acceptable solution; 2. Identifies necessary information and applicable engineering and other knowledge and skills; 3. Generates and formulates possible approaches to solution of problem; 4. Models and analyses possible solution(s); 5. Evaluates possible solutions and selects best solution; 6. Formulates and presents the solution in an appropriate form.
What is Satisfactory Performance?	Satisfactory performance is defined in the rubric in Section 10.3.
What is the consequence of unsatisfactory performance?	The candidate fails the module.

GA 2. Application of scientific and engineering knowledge: Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialisation to develop solutions to complex engineering problems.	
How is the Outcome Assessed?	The candidate must successfully complete an individual project. The project is assessed using a number of reports, oral presentations and poster as stipulated in the due dates section at the start of this document. The candidate: 1. Brings mathematical and numerical analysis to bear on engineering problems by using an appropriate mix of: a) Formal analysis and modelling of engineering components, systems or processes; b) Communicating concepts, ideas and theories with the aid of mathematics; c) Reasoning about and conceptualising engineering components, systems or processes using mathematical concepts. 2. Uses physical laws and knowledge of the physical world as a foundation for the engineering sciences and the solution of engineering problems by an appropriate mix of: a) Formal analysis and modelling of engineering components, systems or processes using principles and knowledge of the basic sciences; b) Reasoning about and conceptualising engineering problems, components, systems or processes using principles of the basic sciences. 3. Uses the techniques, principles and laws of engineering science at a fundamental level and in at least one specialist area to: a) Identify and solve open-ended engineering problems; b) Identify and pursue engineering applications; c) Work across engineering disciplinary boundaries through cross disciplinary literacy and shared fundamental knowledge.
What is Satisfactory Performance?	Satisfactory performance is defined in the rubric in Section 10.3.
What is the consequence of unsatisfactory performance?	The candidate fails the module.
GA 3. Engineering Design: Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs.	
How is the Outcome Assessed?	The candidate designs components, systems, engineering works, products or processes as part of the project. The design process and its outcome is documented in the report. The candidate executes an acceptable design process encompassing the following: 1. Plans and manages the design process: focuses on important issues, recognises and deals with constraints; 2. Acquires and evaluates the requisite knowledge, information and resources: applies correct principles, evaluates and uses design tools; 3. Performs design tasks including analysis, quantitative modelling and optimisation; 4. Evaluates alternatives and preferred solution: exercises judgment, tests implement ability and performs techno-economic analyses; 5. Communicates the design logic and information.
What is Satisfactory Performance?	Satisfactory performance is defined in the rubric in Section 10.3.
What is the consequence of	The candidate fails the module.

unsatisfactory performance?	
GA 5. Use of Engineering Tools: Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	
How is the Outcome Assessed?	Sufficient demonstration of the critical use of applicable engineering methods, skills and tools at the level of 3rd or 4th year BEng is required. The candidate: 1. Uses method, skill or tool effectively by: a) Selecting and assessing the applicability and limitations of the method, skill or tool; b) Properly applying the method, skill or tool; c) Critically testing and assessing the end-results produced by the method, skill or tool. 2. Creates computer applications as required by the discipline.
What is Satisfactory Performance?	Satisfactory performance is defined in the rubric in Section 10.3.
What is the consequence of unsatisfactory performance?	The candidate fails the module.
GA 6. Professional and technical communication: Demonstrate competence to communicate effectively and inclusively on complex engineering activities, both orally and in writing, with the engineering community and society at large, taking into account cultural, language, and learning differences.	
How is the Outcome Assessed?	The candidate demonstrated: The communication was clear and understandable; Oral presentations, poster and final report are professionally acceptable; Language usage is as required for technical communication. The candidate executes effective written communication as evidenced by: 1. Uses appropriate structure, style and language for purpose and audience; 2. Uses effective graphical support; 3. Applies methods of providing information for use by others involved in engineering activity; 4. Meets the requirements of the target audience. The candidate executes effective oral communication as evidenced by: 1. Uses appropriate structure, style and language; 2. Uses appropriate visual materials; 3. Delivers fluently; 4. Meets the requirements of the intended audience.
What is Satisfactory Performance?	Satisfactory performance is defined in the rubric in Section 10.3.
What is the consequence of unsatisfactory performance?	The candidate fails the module. However, if all examiners agree, the student may be given a single opportunity to correct formatting and grammatical errors in the document. No content changes would be allowed. These changes must be made to either the satisfaction of the supervisor or examiners, as agreed on during the examination. These changes will not affect the final grade, but will allow the student one additional opportunity to satisfy GA 6.

GA 8. Individual and collaborative teamwork: Demonstrate competence to function effectively as an individual and as a member or leader in diverse and inclusive teams, and in multi-disciplinary, face-to-face, remote and distributed settings.	
How is the Outcome Assessed?	The candidate demonstrated: The main objectives of the project were achieved; The student's work was focussed on the objectives and well planned; Moderate supervision was required. The candidate demonstrates effective individual work by performing the following: 1. Identifies and focuses on objectives; 2. Works strategically; 3. Executes tasks effectively; 4. Delivers completed work on time.
What is Satisfactory Performance?	Satisfactory performance is defined in the rubric in Section 10.3.
What is the consequence of unsatisfactory performance?	The candidate fails the module.
GA 9. Independent Learning Ability: Demonstrate competence to engage in independent learning through well-developed learning skills.	
How is the Outcome Assessed?	The candidate demonstrated: Applicable independent research was conducted and sensibly used in the project; The source material used was at the level of a 3rd or 4th year text book. The candidate shows evidence of being an effective independent learner by the following: 1. Reflects on own learning and determines learning requirements and strategies; 2. Sources and evaluates information; 3. Accesses, comprehends and applies knowledge acquired outside formal instruction; 4. Critically challenges assumptions and embraces new thinking.
What is Satisfactory Performance?	Satisfactory performance is defined in the rubric in Section 10.3.
What is the consequence of unsatisfactory performance?	The candidate fails the module.

8 Guidelines

8.1 Technical Focus

- The student must ensure that, in the course of the execution of the project, they demonstrates that they can judiciously utilise the technical knowledge gained in the third and fourth year modules to an engineering project, and sufficiently address all the outcomes as required by the module.
- Students must demonstrate in the execution and documentation of the project that the integration of their part of the project in a greater mechanical/mechatronic system was sensibly taken into account.

8.2 Insufficient Progress

- Continuation of the module can be refused at any stage due to insufficient progress. The particular student must drop the module in such a case, or a fail will be awarded.
- All students' progress will continually be assessed by the supervisor. The supervisor can at any stage request the student to do an oral presentation of the project's progress to a panel of three lecturers (appointed by the Chairman; the supervisor is normally one of the panel members). If this panel finds that the student's progress has been insufficient, continuation of the project will be refused.

8.3 Project File and Final Report

- The final report is handed in as a single PDF document via SUNLearn. Additional hard copies may be made at the student's expense for bursary companies, employers, etc. if required.
- Editorial aspects of the final report must be in accordance with the department's "Guide for Writing Technical Reports", which is available on SUNLearn. The font size used in the report MUST be the font size applicable to A4 pages as shown in the writing guide.
- The report may not be more than 50 pages from the introduction to the conclusion. The appendices may not be more than 25 pages. Detail information such as manufacturing drawings, data sheets, etc. must be included in the project file, but not in the report.
- A summary page, in the format given in section 13, must be included in the final report just after the title page (where the abstract would have come). The report must not have another abstract.
- The page following the summary page must contain the Department's plagiarism declaration, available on SUNLearn. The declaration must be dated and signed by the student.
- An outcome assessment must be included in the project proposal, draft and final report directly after the Department's plagiarism declaration page. This assessment is a single page on which the student must indicate which parts of the report presents evidence of achieving each of the ECSA outcomes (as specified in this document). A sentence or two can also be given for each outcome to explain how those parts of the report demonstrate achieving the outcome.
- A summary of all the actual costs incurred it the project as well as a techno-economic analysis must be included as one of the appendices. The costs of the student's own time, purchases, operating costs, the costs associated with use of equipment, etc. must be included.
- A Gantt chart with the project plan as well as the actual execution of the project must be included as one of the appendices.
- A report on the responsible use of resources and an end-of-life strategy must be included as one the appendices.
- A risk assessment of laboratory or experimental setups, including any safety measures/procedures and emergency evacuation procedures relating to experimental work must be included as one of the appendices, typically in the form of a safety report.
- If the objectives of the project changed during its execution, this must explicitly be discussed and motivated in the introduction.
- Do not refer to the report as a "thesis". This is a technical report.
- Even though the final report is the most important document in the determination of the final mark, the final assessment and external moderation will include consideration of the whole

project file, the marks given for the project proposal, progress report, oral presentations, poster and attendance of the research and general lectures.

- Each student must submit a complete project file by the due dates as specified in this document, to pass the module. It is the student's responsibility to compile the project file and to ensure that it is complete. The project file must contain at least the following documents, in the order given here:
    - The original instruction for the project.
    - Every other report submitted (including Project Proposal, Pre-study, preliminary final, etc.), as they were marked by the supervisor.
    - Copies of the weekly progress reports that the student submitted to the supervisor.

8.4 Oral and Poster Presentations

- Each student must on two occasions, as indicated in the due dates section, give oral presentations on their project. The time limit is 10 minutes per presentation. Students being considered for the SAIMechE prizes for best and runner-up presentations must give a further oral presentation.
- The oral presentations are aimed at an audience of graduate engineers. The first presentation is aimed towards presenting the progress of the project, and the second is aimed at convincing the client that the work has been completed and can be invoiced.
- The poster presentation must adhere to the requirements given in Guidelines for Poster Presentations (see SUNLearn) and must be aimed at the general public.
- The poster itself must remain in the department after the presentations.

8.5 Due Dates and Regular Meetings

- All due dates are specified at the start of this document. The due dates for submission of the final reports must be complied with strictly, whether the originally planned work is completed or not. If a student fails due to marks being subtracted due to late submission, they will fail Mechanical/Mechatronic Project and have to do a **new** project with a **new** supervisor in the next year.
- **Extensions for submission of the final report will only be considered if unforeseen circumstances beyond the control of the student lead to a significant delay (e.g. for a few weeks) in the execution of the project. If a student is convinced that they should be granted an extension, they must submit a well-motivated written request to the Departmental Chairman, at least two weeks before the due date. Only a written grant for extension from the Chairman of M&M will be acknowledged for late submission of the final report.**
- Students must meet their supervisors at least **once every second week** in the first semester and **once per week** in the second semester at mutually agreed times. A short report (typically ½ page) on the preceding week's progress and the coming week's objectives, must be handed to the supervisor at each of these meetings. The reports must be kept in the Project File as record of the student's progress.
- All students (**including repeaters**) must attend the scheduled lectures. All lectures are compulsory.

8.6 Study Guidance

- The student is responsible for their whole project, as for any other subject. They must do their best, with the outcomes of the module in mind, in the time allocated to the module and report the work in the final report.
- The supervisor will guide the student in the formulation of the objectives for the project.
- The supervisor will give reasonable, but restricted guidance during the execution and reporting of the project.
- The supervisor will review the draft of the final reports in detail and give the student feedback on the quality of the work and the report writing.

8.7 Assessment

- Note section 9.6 gives the point of departure for the assessment.
- The project proposal, progress report, final report, project file and poster represent the examination scripts.
- The supervisor and an internal moderator evaluate (independently of one another) the student's work and their reporting based on the final report, oral presentations and poster. They may also consider the project file. Marks for the technical content and the complementary content are awarded separately. A student must pass both sections, as well as each ECSA outcome, to pass the module.
- External examiners for the module are nominated by the Department's Executive Committee, and are then approved by the Faculty of Engineering's Faculty Board. The external examiners are graduate engineers with appropriate industry experience.
- Each student's final report and marks are then considered by at least one external examiner. The reports are sent to the external examiners several weeks before the oral presentations. The examiners visit the Department during the November examination period, during which the oral and poster presentations will take place. Each student will deliver a 10 minute oral presentation, after which they will be questioned by the supervisor, internal moderator and the external examiner. The internal moderator will facilitate the process of awarding the final mark, based on the student's defence of their work, and the marks awarded by the supervisor and external examiner. The external examiners will also consider the grades given for the project proposal, progress report and progress presentation, and the attendance record of the research lectures will also be taken into account for assessment.
- The marks approved by the external examiners are final. No re-marking will be allowed because the projects had already been subjected to external examination.

9 Procedures

9.1 General

- Students must note that specific procedures must be followed for the following:
    - General safety instructions.
    - To have something made in the workshops.
    - To have something 3D printed at the department.
    - To have laser cutting performed outside of the department.

- For the purchase of equipment or components.
- For the use of the laboratories and instrumentation store.
- For the use of any equipment off site.

- Projects that are led by the Department of Mechanical and Mechatronic Engineering are funded by Mechanical and Mechatronic Engineering and are subject to the procedures of that department. Similarly, projects that are led by the Department of Electrical and Electronic Engineering are funded by Electrical and Electronic Engineering and are subject to the procedures of that department.
- Please refer to SUNLearn for the procedures applicable in the Department of Mechanical and Mechatronic Engineering (including specific procedures for the above mentioned items), and to http://www.ee.sun.ac.za for the Department of Electrical and Electronic Engineering's procedures.

9.2 Safety

- Students must study the general safety instructions and complete the attached indemnity form, before he/she may enter the laboratories. All students must attend the safety training lecture.
- Students are warned to work safely and to adhere to the general safety instructions.
- Please contact the technician in charge of the lab, lecturer or departmental chairperson, should you become aware of any dangerous situation in the labs.

10 Plagiarism

Plagiarism is viewed as a very serious offense and will not be tolerated. “Turnitin” will be used to check all reports and will automatically report instances of plagiarism.

Plagiarism will be treated in accordance to the University's policy for the prevention and handling of plagiarism (explained in the Calendar Part 1). This policy defines plagiarism as the theft and use of the ideas, material and other intellectual property of others that are passed off as one’s own. Plagiarism therefore includes presenting another student's work as your own, as well as to use information created by someone else without expressly and unambiguously acknowledging the source.

Note: In the preliminary reports, supervisors will not necessarily bring their suspicions of plagiarism to the attention of students.

11 Marking Schedules

Please note that all marking schemes are just examples and may change during the year

11.1 Assessment Scheme for Project Proposals

Student					Supervisor								
Name: Student No:					Name: Signature:								
Outcomes[1]	**Fail**				**Pass**					**Distinction**			
Background and motivation	• Background and motivation is not clearly communicated. • Project purpose and context is not clear.				• Clear, concise background and motivation. • Project purpose and context are clear to readers not familiar with the project.					• Purpose and context of project is based on sound research. • Conveys familiarity with the study area/topic.			
	2	4	6	8	10	11	12	13	14	15	16	18	20
Problem statement and objectives	• Problem statement is not clear. • Objectives are vague and/or have superfluous detail. • Objectives are generic in detail.				• Concise problem statement and objectives. • Problem statement and objectives are clear to readers that are not familiar with the project. • Clear statement of end deliverable.					• Problem statement and objectives are at the level expected from a person eminently familiar with the study area/topic. • Difficult concepts are explained such that readers less familiar with the topic can understand the goal and challenges.			
	2	4	6	8	10	11	12	13	14	15	16	18	20
Planned activities, deliverables and risks	• Significant omissions in the steps or superfluous detail. • Little/no attention to risks. • Vague descriptions of steps. • Unrealistic scheduling.				• Due consideration to scheduling, available facilities and risks. • Almost all steps are shown and with enough detail that it is clear what will be done. • Deliverables are described for each step.					• Provide supporting information to a level of detail expected of a person eminently familiar with the study area/topic. • All significant steps are shown. • Project risks are properly addressed and risk mitigation is proposed.			
	2	4	6	8	10	11	12	13	14	15	16	18	20
Responsible use of Resources	• Superfluous detail given or major omissions • Resource use plan bear little/no relation to project				• Key project elements are covered by the resource use plan • Key project elements clearly identified • Reasonable resource use plan for project at hand					• Resource use plan at a level of detail expected of a person eminently familiar with the study area/topic • Clear end-of-life strategy			
	2		4		5		6		7	8		10	
Budget	• Superfluous detail given or major omissions. • Budget and project plan bear little/no relation.				• Most project (project plan) and budget elements (material, labour, equipment) are covered. • Reasonable cost estimates, with supporting information as relevant. • Required equipment / materials clearly identified					• Equipment and material identified to a level of detail expected of a person eminently familiar with the study area/topic. • Due consideration of available equipment.			
	2		4		5		6		7	8		10	
Professional and technical communication	• Communication is unclear and not easy to understand. • The report is not to a professional standard.				• Communication is clear and understandable. Report is professionally acceptable. • Language usage is as required for technical communication.					• The report is brief and to the point. Difficult concepts were conveyed so that it was understandable at first reading. • The material was compiled systematically so that the arguments flow logically.			
	2	4	6	8	10	11	12	13	14	15	16	18	20
										Final Mark:			

[1] Note: These outcomes are not the same as the Final Year Project ECSA outcomes. They pertain only to the Project Proposal.

11.2 Assessment Scheme for Project Progress Oral Presentation

Assessment Scheme for Oral Presentations for Mechanical and Mechatronic Projects

- The focus of this evaluation form is the presentation skills of the student
- Feedback on the progress to date and planned activities for the remainder of the project is important

Student Name: ……..……..……..…..

Student Number: ……..……..……..…..

VERBAL COMMUNICATION (audibility, voice tone & quality, tempo, language use)	20	18	16	14	12	10	8	6	4	2
NON-VERBAL COMMUNICATION (enthusiasm, posture, eye contact, gestures, facial expression)	20	18	16	14	12	10	8	6	4	2
CONTENT (credibility, technical content, conclusions, comprehensibility)	20	18	16	14	12	10	8	6	4	2
MEDIA USE	10	9	8	7	6	5	4	3	2	1
SAFETY AND RESOURCE USE (risks identified, safety procedures in place, resource use plan in place)	10	9	8	7	6	5	4	3	2	1
OVERALL IMPRESSION	20	18	16	14	12	10	8	6	4	2

TOTAL _____%

NOTE: **The time for each presentation is 10 minutes. Students are penalised if they exceed this limit.**

11.3 Assessment Scheme for Final Examination

Student	Examiner/Moderator
Name: Student No:	Name: Signature Role (circle the suitable choice): Supervisor / Internal / External

Mark each outcome's level with an X in the Fail, Pass or Distinction column
NB: A fail in ANY outcome leads to a final mark of less than 50

Outcomes	Fail	Pass	Distinction
Problem Solving	The student made an attempt, but problem identification or solution was unsatisfactory.	A complex engineering problem has been identified and solved systematically. Engineering judgement was sufficiently exhibited.	A complex engineering problem has been effectively solved and the results realistically interpreted. The interpretation/solution is innovative or advanced.
	3 6 9	10 11 12 13 14	15 16 18 20
Application of Scientific and Engineering Knowledge	Basic knowledge was used, but not correctly or the technical depth is not sufficient. No coherent and/or critical application of knowledge is demonstrated.	Technical depth is at the 3rd or 4th year of the BEng programme. Coherent and critical application of the fundamental principles and theory were adequately demonstrated.	Knowledge was applied to the problem in an innovative way. In exceptional cases, technical depth is advanced.
	3 6 9	10 11 12 13 14	15 16 18 20
Engineering Design *(including design of an investigation, experiment or software)*	An attempt has been made to design something, but the result was not of significant value.	Alternatives were critically considered. It was demonstrated that the project's result is useful and utilises knowledge from the applicable areas.	The project's results can be utilised as is, has been thoroughly completed and/or is of significant practical or research value. Complex systems and/or advanced, custom peripherals were developed and critically investigated.
	2 4 6	8 9 10 11	12 13 14 15
Engineering Methods, Skills and Tools *(including IT)*	Applicable methods/tools were not used or were used incorrectly. The techno-economic analysis and/or resource use plan is missing.	Demonstrating the critical use of applicable engineering methods, skills and tools at the level of 3rd or 4th year BEng. The techno-economic analysis and resource use plan is acceptable.	In addition to the successful use of the basic tools, new tools (including more advanced use of FEM, CFD or other engineering software packages) were mastered and applied to the task.
	2 4	6 7 8	9 10 11 12
Professional and Technical Communication	Objectives, problem statement, content or scope of the project not clearly communicated, or the oral presentations, reports or poster were not of a professional standard.	The communication was clear and understandable. Oral presentations, poster and final report are professionally acceptable. Structure, style, and language are as required for technical communication, and according to Departmental guidelines.	The student gave a concise explanation of a large volume of work. Difficult concepts were conveyed so that it was understandable at first reading. The material was compiled systematically so that the arguments flow logically.
	2 4 6	7 8 9	10 11 12 13
Individual Work *(After discussion with supervisor)*	A significant part of the project objectives were not achieved due to, e.g. poor planning, time management, or interaction with co-workers. A lot of supervision was required.	The main objectives of the project were achieved on time. The student's work was focussed on the objectives and well planned. Moderate supervision was required.	The student showed initiative and generated innovative ideas that improved the project's results beyond the original objectives. Student required little supervision.
	2 4	5 6 7	8 9 10
Independent Learning Ability	New information was gathered, but applied incorrectly or incompletely. Too little evidence of the essential research required for the project.	Applicable independent research was integrated in project plan, conducted critically and sensibly used in the project. The source material used was at the level of a 3rd or 4th year text book.	In exceptional cases, the research conducted by the student or the level of difficulty of the work is at or near post-graduate level. The student read widely and used the source material critically.
	2 4	5 6 7	8 9 10

Final mark	**%**

11.4 Moderation Guideline for Final Mark Allocation

Mark	Project Manage-ment	Task Completed	Did more than required	Did better than required	Original work (publishable)	Presentation
90+	Perfect	Perfect	Yes	Yes	Yes	Perfect
75+	Excellent	Excellent	Yes	Yes	No	Excellent
70+	Good	Good	Yes	Yes	No	Good
60+	Average	Yes	Yes/No	No/Yes	No	Acceptable
50+	Acceptable	No, but …	No	No	No	Acceptable
40+	Poor	No	No	No	No	Acceptable
39-	Poor	No	No	No	No	Unacceptable

12 Prescribed Summary page

Note: the wording in the table will differ slightly for the Project Proposal and for the Final Report. Keep the appropriate wording in each report and omit the remainder.

Executive Summary

Title of Project
Objectives
What is current practice and what are its limitations?
What is new in this project?
If the project is successful, how will it make a difference? *(For the final report please change this heading to: "Was the project successful? How will it make a difference?")*
What are the risks to the project being a success? Why is it expected to be successful? *(For the final report, please change this heading to: "What were the risks to the project being a success? How were these handled?")*
What contributions have/will other students made/make?
Which aspects of the project will carry on after completion and why?
What arrangements have been/will be made to expedite continuation?

13 General Safety Instructions

UNIVERSITY OF STELLENBOSCH
DEPARTMENT OF MECHANICAL AND MECHATRONIC ENGINEERING

- **Help prevent risks to people and machines by being careful, alert and pro-active.** When in doubt, ask the technician or lecturer in charge.
- **According to the Public Health and Safety Act (no 85 of 1993), a student is considered an employee of the University of Stellenbosch.** Copies of the Public Health and Safety Act, no 85 of 1993 are available from the Departmental Chairman.
- **If a student is instructed by either his/her supervisor, another lecturer or technician in with regards to safety, the Act prohibits him/her from disobeying such instructions. The Act does not recognise a statement such as "at their own risk".**
- **These instructions are applicable to students, personnel and persons entering the laboratories and other rooms of the Department of Mechanical and Mechatronic Engineering.**

PLEASE NOTE – The following information may be updated throughout the year.

In case of an emergency: Contact the technician/lecturer/Campus Security **Emergency phone:** located in the entrance hall of the laboratory block/MMW entrance				
DIVISION	**TECHNICIAN / LECTURER IN CHARGE**	**ROOM No.**	**WORK TEL No.**	**HOME TEL No.**
General	Prof Johan van der Spuy Mr Cobus Zietsman *CAMPUS SECURITY*	M5006 M219	3742 4275 4666	083 456 2844 083 767 1835 4666
Thermo-fluids	Mr Cobus Zietsman Prof Mike Owen	M219 M5033	4275 4266	083 767 1835 083 456 2844
Structural Mechanics	Mr Nathi Hlwempu Prof Martin Venter	M141 M6037	4488 4477	083 735 6715 082 866 0836
Mechatronics	Dr Danie Els Mr Kevin Neaves	M5006 M2114	4248 4279	082 474 4891 072 109 5926
Workshop	Mr Ferdi Zietsman	Workshop M1109	4954	083 233 1646
Fire Brigade			887 1333	
Ambulance			883 3444	
Police			10111 of 883 3222	
Student Health Services			808 3496	

PERSONNEL	FIRST-AID TRAINED	TEL No	ROOM No	FIRST-AID KIT
Cobus Zietsman	Yes	4275	M219	No
Simone Paulse	Yes	4278	MMW/Store/106/214	Yes
Graham Harmse	Yes	4278	M106	Yes
Julian Stanfliet	Yes	4488	M142	Yes
Kevin Neaves	Yes	4279	M3001	Yes
Gillian Cortereal	Yes	4374	M5005	Yes
Shawn Hendrikse	No	9976	M5010	Yes
Nathi Hlwempu	Yes	4488	M141	Yes

13.1 All Hours

- **No person is permitted to work alone in a laboratory.** When a student has permission to work after hours, another person should always be present, regardless of the nature of the work.
- **Do not enter the Internal Combustion Engine Laboratory while the fire alarm is switched on.** The automatic fire-fighting system release gasses which can lead to suffocation.
- **After hours emergencies:** Contact Campus Security (Tel. 4666).

13.2 Reporting of safety risks and incidents

- Report all safety risks, for example unsafe/defective electrical wiring and cables, air-pressure pipes, *etc.* either to the technician/lecturer in charge or to the Departmental Chairman.
- Report all incidents of injury to persons either to the technician/lecturer in charge or to the Departmental Chairman.

13.3 Dress & Safety Equipment

- Closed shoes should always be worn in the laboratories. Bare feet or open toed shoes are not allowed.
- Safety shoes must be worn when heavy equipment is being moved.
- Eye protection safety equipment must be used where eye injuries could possibly occur.
- Ear protectors should be used in case of moderate to high noise levels.
- All persons should be aware of the location of fire extinguishers or fire hoses as well as emergency exist before you start working.
- Loose clothing or long hair is not allowed near moving machines.
- Any lab specific requirements should be followed.

13.4 Permission required

- **No person is allowed to handle dangerous equipment on his/her own.** Dangerous equipment refers to any equipment with rotating or moving parts or equipment that can cause any other danger such as explosions, fire or electrocution. At least one other person must be present in the laboratory when such

dangerous equipment is used. When a student has permission to work after hours, another person should always be present, regardless of the nature of the work.

- **Only authorised personnel are allowed to operate electrical equipment.** The Act is very strict in this instance and a student is not even allowed to connect a 15 A plug.
- **Students should first acquire permission before they work in any laboratory or with any equipment.** Permission should be obtained from the person in charge of the particular equipment or laboratory. The names of the relevant personnel are clearly stated on the doors of the laboratories.

13.5 General safety regulations applicable to laboratories

- Smoking is not allowed in either the laboratories or the workshops.
- No one is permitted to work in a laboratory after consuming alcohol.
- The laboratories should be cleaned every day after the completion of any work.
- Persons working in laboratories where gases and fumes can build up must personally ensure that the ventilation is adequate.
- Fuels, paints and other flammable materials or materials that give off fumes, should be stored in the correct manner.

13.6 Misuse of Equipment and Negligence

- **Any student that fails to return equipment to the store according to the store's rules, will be fined R100.**
- If equipment must be replaced or repaired as a result of negligence by a student, the student must contribute towards the costs. The student's contribution will be calculated as follows:
    - If the replacement / repair cost of the equipment is R300 or less, the student will be held responsible for the entire amount.
    - If the replacement / repair cost of the equipment in excess of R300 the student must pay one third of the insurance excess fare, or a third of the equipment's replacement / repair value if the insurance does not accept the claim.
    - The student must pay the fine at Admin into cost centre 368F and must then show the receipt to the laboratory manager before he/she will be allowed to use departmental equipment again. The decision of whether a fine is applicable to the student, will be with the laboratory manager, but the student may appeal the decision with the Departmental Chariman.

13.7 Off-site use of Equipment

- **All equipment that will be used off-site, must be booked out before hand. Students will be fully responsible for any damage to or theft of equipment that was not properly booked out before hand.**